MERIDIAN JUNIOR COLLEGE
Preliminary Examinations 2011
Biology H1 8875/2 – Mark Scheme

1(a) Describe how DNA is packaged in eukaryotic cells during interphase. [3]

- In eukaryotes, **negatively charged DNA** wrapped around **positively charged** proteins called **histones via ionic bonds**.
- to form **nucleosomes**
- **Linker DNA** joins **the nucleosomes**.

(b) DNA replication occurs in cells during interphase before they divide by mitosis. Explain why it is important that replication occur before mitosis. [3]

- Replication of DNA results in the **DNA content of the cell being doubled**
- This is to ensure that **each daughter cell** has exactly the **same genetic material as** the **parent cell after mitosis** and cytokinesis.
- Maintain the **diploid state** of the daughter cells

(c) State **two** differences in the behaviour of chromosomes in mitosis compared to meiosis. [2]

Mitosis	Meiosis
1. Homologous chromosomes do not associate during prophase	**1. Homologous chromosomes associate to form bivalents in prophase I**
2. Chiasmata are never formed	**2. Chiasma are likely to form**
3. Crossing over never occurs	**3. Crossing over likely to occur**
4. Chromosomes form a single row at the equator of the spindle during metaphase	**4. Chromosomes form 2 rows at the equator of the spindle during metaphase I**
5. Chromatids move to opposite poles during anaphase	**5. Chromosomes move to opposite poles during anaphase I**

Some mothers of infants with Down's syndrome have a mutation of a gene coding for an enzyme which is involved in the metabolism of methyl groups. A lack of methyl groups on DNA results in mistakes in the segregation of chromosomes during cell division and in structural mutations of chromosomes. Individual with Down's syndrome has an extra chromosome 21.

(d) Explain how mistakes in segregation resulted in Down's syndrome. [2]

- **Non-disjunction during meiosis**
- **Failure of chromosome 21 to separate/ both end up in the same daughter cell**
- **When abnormal gamete fuses with normal gamete, resulted in extra copy of chromosome of 21/ 2n +1;**

[Total: 10]

2. Fig. 2.1 and Fig. 2.2 are diagrams that show the main details of 2 processes.

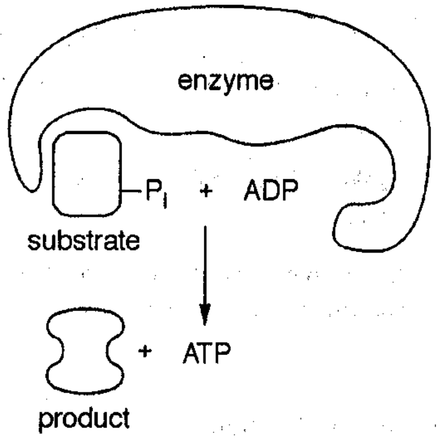

Fig. 2.1

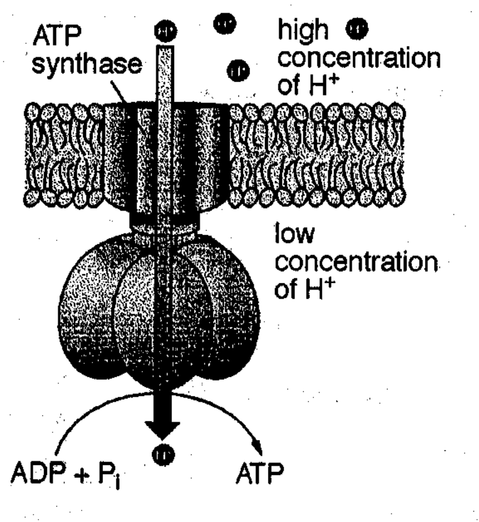

Fig. 2.2

(a) State precisely where these two processes occur in a cell. [2]

Substrate level phosphorylation : **Cytoplasm/ mitochondrial matrix**
Oxidative phosphorylation : **Inner mitochondrial membrane**

(b) Only substrate level phosphorylation is possible in the absence of oxygen.
Explain why oxidative phosphorylation is not possible in the absence of oxygen. [3]

- **In the absence of oxygen, the final electron acceptor [1],**
- **the electron transport chain cannot function / electron carriers are not re-oxidized [1].**
- **Oxidative phosphorylation does not take place as the proton gradient cannot be established for ATP synthesis [1]**

Fig. 2.3 shows a protein complex found on the membrane of cells lining the stomach. An inhibitor acts by irreversibly blocking an enzyme in the protein complex. It is found that this inhibitor helps to reduce gastric acid secretion in the stomach.

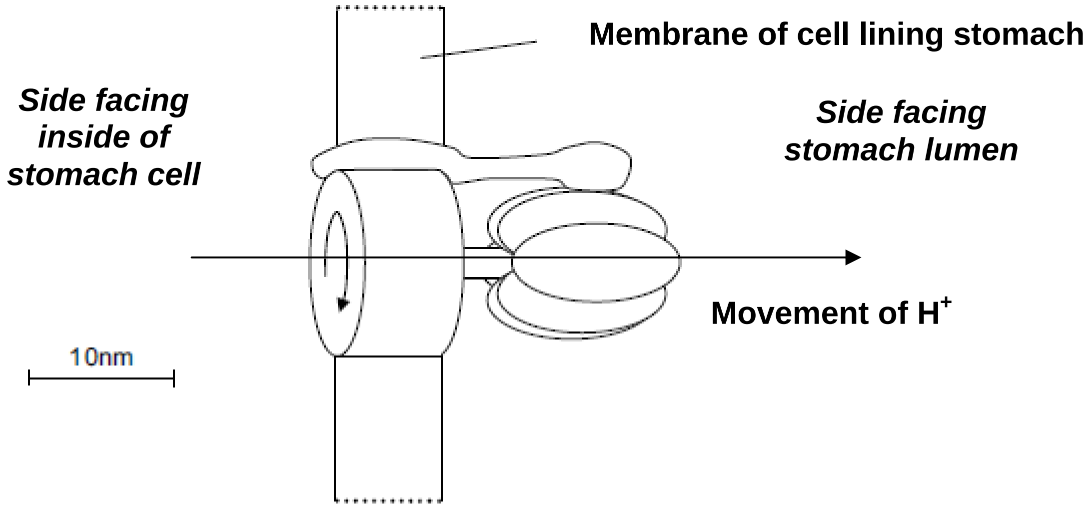

Fig.2.3

(c) Explain how the inhibitor helps to reduce the secretion of acid into the stomach. [2]

- The inhibitor binds **covalently** to the **ATP synthase** in the **proton pump**.
- The inhibition disallows the **diffusion of hydrogen ions** through the stomach membrane into the stomach lumen, thus, reducing the acid secretion.

Researchers found out that these inhibitors could induce the apoptosis of gastric cancer cells.

(d)(i) Explain what is meant by the term 'cancer'. [3]

- cells **fail to respond to normal controls** on their multiplication and enlargement
- **divide excessively** / **uncontrollably** by mitosis
- cells are **unspecialized**
- forming **tumours**

(ii) Name **two** factors that would increase the chances of the normal cancer gene becoming mutated. [2]

- **Exposure to carcinogens.**
- **Infection by viruses.**
- **UV radiation.**
- **Increasing age.**

(iii) State one similarity between cancer cell and stem cell. [1]

- Both replicate indefinitely
- Both are undifferentiated
- Both can be found in various parts of the body.

[Total: 13]

3. Flax may be attacked by different races of a fungus, *Melamspora lini*, causing flax rust. One variety of flax, Ottawa, is resistant to race **24** of *Melamspora lini*, but is susceptible to race **22**. Another variety, Bombay, is susceptible to race **24** of *Melamspora lini*, but resistant to race **22**. When the two varieties of flax are interbred, the F_1 generation is resistant to both races of rust. One investigation into selfing the F_1 gave the following results:

128 plants resistant to both races **22** and **24** of rust,
39 plants resistant to race **22** but susceptible to race **24** of rust,
44 plants susceptible to race **22** but resistant to race **24** of rust,
14 plants susceptible to both races **22** and **24** of rust

(a) Using the given symbols, state, with reasons, the genotypes of the two parental varieties of flax. [4]

Symbols:	**A**	- allele conferring reistance to race 22 *M. lini*
	a	- mutant allele; flax susceptible to race 22 *M.lini*
	B	- allele conferring resistance to race 24 *M.lini*
	b	- mutant allele; flax susceptible to race 24 *M.lini*

Genotypes:	variety Ottawa	**aa BB**
	variety Bombay	**AA bb**

[1]

Reasons:

- F1 are **all resistant to both races of *M. lini*** therefore parental varieties are **double homozygous**
- F2 has **9:3:3:1 ratio** is typical of **dihybrid inheritance** of alleles (2 gene loci) showing **complete dominance**
- The **alleles of the 2 genes segregate independently**
- Susceptibility to each race of fungus is due to the presence of **2 recessive alleles at the respective gene loci**.

[3]

The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**. Complete colour blindness, unlike red-green colour blindness, is controlled by a different gene which is not on the X chromosome. The allele for normal vision, **B**, is dominant to the allele for complete colour blindness, **b**.

Fig. 3.1shows the phenotypes of members of a family in which both types of colour blindness occur.

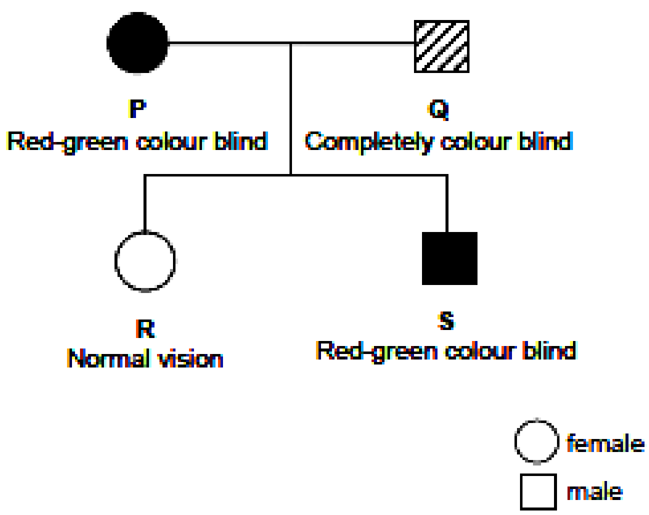

Fig. 3.1

(b) Using genetic diagrams and the symbols given, show clearly the cross between individuals **P** and **Q**. [4]

Parental phenotypes: Red-green colour blind Mother x Completely colour blind father

Parental genotypes: BB X^g X^g x bbX^GY [1]

Gametes produced by parents: BX^g BX^g x bX^G bY [1]

Genotype of offspring: Bb X^GX^g Bb X^GX^g Bb X^gY Bb X^gY [1]

Phenotype of offspring: Normal vision, female Red-Green colour blind male

[1]

Phenotypic ratio: 1 : 1

It was found that the complete colour blindness is due to a mutant allele on the chromosome. A normal allele codes for the development of normal cones (pigments cells in the retinal layer of the eye)

(c) Suggest how this mutant allele leads to abnormal development of cone cells. [1]

- **Loss-of-function mutation/ Mutation in DNA results in production of non-functional protein/no protein production;**

[Total: 9]

4. The human insulin gene coding for insulin is inserted into a plasmid vector. Fig. 4.1 shows the plasmid vector. The polylinker contains 10 different restriction sites.

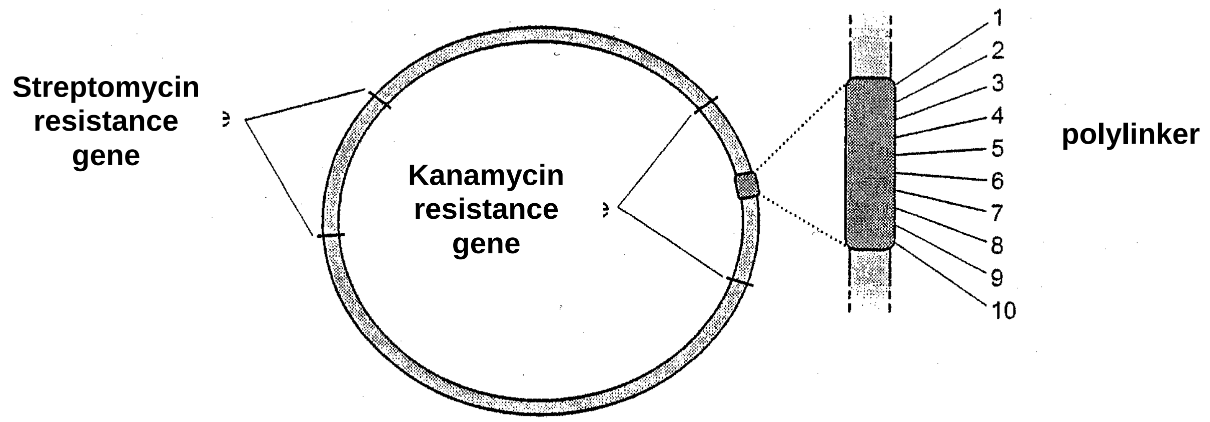

Fig. 4.1

(a) With reference to Fig. 4.1, explain the advantage of the plasmid having two antibiotic resistance genes and a polylinker within the kanamycin resistance gene. [3]

Polylinker:

- **Different restriction enzymes** can be used for cloning

2 antibiotic resistance genes:

- This allows for the **insertion of gene of interest**
- To identify the bacterial colonies containing the **recombinant plasmid**

(b) State a problem associated with cloning a eukaryotic gene into a prokaryotic plasmid vector and suggest a solution to overcome the problem. [2]

Problem

- **Presence of introns** in eukaryotic genome gives rise to **non-functional protein** when expressed using prokaryotic system

OR

- **Presence of introns** in eukaryotic genomes that **cannot be removed by the prokarytotes**

Solution

- Use **cDNA** synthesize from mRNA via **reverse transcriptase**

(c) Discuss ethical issues raised by the use of genetically modified organism. [3]

- There is little concern about whether the animals are biologically suited to the use of growth hormone to withstand the additional burden of increased meat production.
- Plants that have been modified to carry animal genes may be a concern for the vegetarians or certain religious groups.
- Cloning raises the possibility of breeding many identical copies of animals. The concern is that the techniques will be used to clone humans.

[Total: 8]

Section B [20 marks]

5(a) Describe the main ways in which a globular protein differs from DNA. [7]

Feature (11 mk pts)	**Globular protein**	**DNA**
Monomers [1] [1] [1] [1]	Amino acids 20 different types	Deoxyribonucleotides, comprising phosphate group,nitrogenous base and deoxyribose 4 different types Adenine, thymine, guanine, cytosine
Bonds between monomers [1]	Peptide bonds	Phosphodiester bonds
Hydrogen Bonding within molecule [1] description + [1] complementary	Hydrogen bonds between C=O and N-H groups of peptide bond regions Hydrogen bonds between R groups of amino acids	Hydrogen bonds between complementary bases AT, GC or purine/pyrimidine
Other bonds [1]	Globular / tertiary structure maintained by hydrogen bonds, disulphide bonds, ionic bonds and	Hydrogen bonds between complementary bases and hydrophobic interactions

	hydrophobic interactions between R groups	between stacked bases
No. of chains/strands [1]	May be a single polypeptide chain or two or more polypeptide chains	Double stranded / double helix
Foldings [1] + [1]	Primary sequence folds upon itself to give secondary structure [1], which further folds into tertiary structure [1] *Reject : "three dimensional shape" as coiling of DNA also into 3D structure*	Linear or circular Double helix may be coiled around histones [1] and scaffold proteins to form condensed chromatin threads/ chromosomes [1]

(b) Describe the role of messenger RNA in protein synthesis. [8]

1. The genetic message for synthesis of a polypeptide is present on the **DNA** in the nucleus.
2. The mRNA molecule is **formed from transcription** of the specific region of DNA which codes for the polypeptide.
3. It acts as a **carrier molecule, conveying the message from the nucleus to the ribosomes in the cytoplasm**
4. Each mRNA molecule has a **sequence of nucleotides**.
5. Each **triplet of bases/codon** codes for **one amino acid**.
6. **mRNA codons specify the order in which amino acids are sequenced** to form a polypeptide
7. The **start codon or AUG on the mRNA** **initiates translation** coding for methionine
8. and the **stop codon** (UAG, UAA, UGA) **terminates translation**
9. **codons on mRNA** matches, via **complementary base pairing**, the **anti-codon in tRNA which carries a specific amino acid**
10. a **single mRNA can be attached by many ribosomes** (polysomes) to synthesise **multiple copies of the same polypeptide**, thus increasing speed of translation

(c) Discuss if DNA cloning can be considered as a form of evolution. [5]

1. Cloning involves **altering the genes of an organism**
2. Under **favorable environments/selection pressures**, the transgenic organisms with the altered genes may be **at selective advantage**
3. **Able to survive and reproduce to pass on these alleles to the offspring**
4. **Over time, the frequency of alleles that give rise to the selective advantageous traits increased.**

5. **If <u>sufficient differences accumulate between isolated populations,</u> new species may arise when members of two populations no longer interbreed to produce fertile offspring. / divergent evolution**

6. **DNA cloning can be considered as a form of evolution**

6(a) Compare the structure and function of *Taq* DNA polymerase and RNA polymerase. [7]

	***Taq* DNA polymerase**	**RNA polymerase**
Similarities:		
1. Template	Both use DNA as template	
2. Structure	Both are proteins/enzymes	
Differences:		
3. Product	DNA	RNA
4. Substrate	Uses deoxyribonucleotides as substrates	Uses ribonucleotides as substrates
5. Template	Uses both DNA strands as template	Transcribes only 1 of the 2 DNA strands
6. Molecular structure	Large no. of disulphide bonds to allow *Taq* DNA polymerase to withstand high temperature	Does not have large no. of disulphide bonds, hence, denatures at high temperature
7.	Requires the presence of free 3'OH group to add the deoxyribonucleotides	Does not requires the presence of free 3'OH group to add the ribonucleotides
8. Functions	Involves in DNA replication/ PCR	Involves in protein synthesis

(b) Outline how restriction enzymes are used for removing sections of DNA from a chromosome. [5]

1. Each restriction enzyme recognizes a **specific sequence of 4 – 8 nucleotide bases** on the DNA molecule/ **<u>restriction site</u>**.

2. Restriction enzyme has **specific active site** that can only **bind with the complementary shape of restriction site**

3. The restriction enzyme acts on the restriction site by **breaking the phosphodiester bonds** at a position between 2 specific bases

4. cutting up DNA molecules into **<u>restriction fragments</u>**.

5. Restriction enzymes produce restriction fragments with **<u>sticky ends or blunt ends</u>**

(c) Explain how gel electrophoresis can be used to provide evidence of molecular homology. [8]

1. Gel electrophoresis can be used in **RFLP analysis** or **protein analysis** to provide evidence of molecular homology.

2. **Different samples of DNA from different species** are taken and **digested/hydrolysed with a restriction enzyme**

3. that breaks the phosphodiester bonds **at specific restriction sites.**

4. Species that **differ genetically** will produce **DNA fragments of different length/size** and vice versa / are **genetically similar** will produce **DNA fragments of similar length/sizes**.

5. The DNA fragments are **loaded into the gel** and subjected to an **electric field**.

6. As DNA is **negatively charged**, the fragments will move towards the anode

7. and they will be **separated according to molecular size** as the **longer fragments will travel more slowly** through the gel/ the **shorter fragments will travel faster** through the gel.

8. The band pattern can be **visualised** by staining the gel with **ethidium bromide/other DNA staining dyes** and **viewing it under UV light**.

9. The **band pattern** produced can be **analysed** / **used to compare samples** from different species in order to establish molecular homology.

10. The **more similarities between 2 species** in the band pattern produced, the **more closely related the species** are and vice versa./ the **more recent they share a common ancestor**

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